UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY

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KCPM - 21KTPM2 - CASE STUDY (PROJECT)

COURSE: SOFTWARE TESTING

Test Summary Report

09/06/2024 – 27/08/2024

COURSE CODE: CSC13003
Group: 11
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Ho Chi Minh City – 2024

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PROJECT INFORMATION

Course Code: CSC13003

Course Name: Software Testing

Title: CS423 – CSC13003 – Software Testing CASE STUDY (PROJECT)

Topic: Test Summary Report

Format: Each student is asked to work on different product features.

The case study requirements will be divided into weekly individual assignments (Requirement 2, 3, 4, 5, 6, 7, 8) and group assignments (Requirement 1, 9, 10). For individual assignments, each student is asked to write their own reports. For group assignments, just one group report is required for each group assignment. However, in the final submission, groups must organize all their work as follow:

Description:

Software Under Testing: Test Management Website, currently in development with planned releases as follows:

- Prototype: Static web interfaces showcasing the main business functions.
- Build 1: Dynamic system retrieving information from a database, focusing on information display.
- Build 2: Complete system enabling users to manage the software testing process.

1. **Requirement 1: Test Plan:** Create a comprehensive test plan using the provided template, customized to meet the project's needs.
2. **Requirement 2: Domain Testing:** Apply domain testing techniques to design and execute test cases for at least 3 features, ensuring thorough coverage and bug identification.
3. **Requirement 3: State Transition Testing:** Use state diagrams to design and execute test cases for key features, ensuring that each state is tested comprehensively.
4. **Requirement 4:** Automation Testing: Automate the test cases from the domain testing phase using tools like Selenium or Katalon, focusing on data-driven testing and verification techniques.
5. **Requirement 5: GUI Testing:** Perform GUI testing using a checklist that covers a minimum of 30 items, ensuring the interface meets quality standards.

6. **Requirement 6: Performance Testing:** Conduct load, stress, and spike testing using tools like JMeter, and document the results with detailed reports and recorded videos.
7. **Requirement 7: API Testing:** Test at least 3 APIs with a minimum of 10 test cases each, focusing on thorough validation of API functionality.
8. **Requirement 8: Data Generation:** Use data generation tools to create meaningful test data, including users, projects, test cases, and bugs.
9. **Requirement 9: Test Case and Bug Tracking:** Log all test cases and bugs in a tracking system, ensuring that all testing activities are properly documented and traceable.
10. **Requirement 10: Test summary report:** Groups must write a test summary report for the entire testing process.

Faculty Instructors: Ms. Trần Thị Bích Hạnh, Mr. Trần Duy Hoàng, Mr. Trương Phước Lộc, Mr. Hồ Tuấn Thanh

GROUP INFORMATION

Group: 11

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21127360	Phạm Văn Minh	pvminh21@clc.fitus.edu.vn	
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TEST SUMMARY REPORT

I. Purpose:

The Test Summary Report is intended to provide a comprehensive overview of the testing activities and results for the Test Management Website project. This report will cover both Build 1, Build 2, Build Final detailing the testing process, including the design and execution of test cases, the outcomes of these tests, the identification and reporting of bugs, and any challenges encountered during the testing phases. The goal is to offer a clear and thorough understanding of the project's quality and readiness for deployment.

II. Application Overview:

The Test Management Website is a web-based application developed to facilitate the management of software testing processes. The application is designed to support users in various testing-related activities, including test planning, test case management, bug tracking, and reporting. Key features of the application include creating and managing test plans, executing and documenting test cases, logging bugs, and generating test reports, all within an integrated platform.

III. Testing Scope:

1. *In Scope:*

Build 1- The testing for Build 1 includes the following areas:

- **Register Account:**
  + **Functionality:** Users can register an account by providing a login name, full name, password (hashed before storing in the database), and avatar.
  + **Constraints:** Test cases are designed to ensure that the registration process works correctly, including validation of required fields, password hashing, and successful storage of user data.
- **Login:**
  + **Functionality:** Users can log in using their login name and password.

    + **Verification:** Testing includes verifying successful logins, handling incorrect credentials, and ensuring that tokens are issued correctly.
- **Logout:**
    + **Functionality:** Users can log out of the system, invalidating their active session.
    + **Constraints:** Test cases confirm that logging out correctly invalidates tokens and ends the user session.
- **Token Management:**
    + **Functionality:** The system generates and issues JWT-based access tokens and refresh tokens upon user request.
    + **Constraints:** Tests ensure that tokens are issued securely and function as expected in securing subsequent API requests.

Build 2- The testing for Build 2 covers the following enhancements and new features:

- **Token Management:**
    + **Functionality**: The system reinitializes access tokens for users when requested.
    + **Verification**: Test cases confirm that access tokens are correctly refreshed and that expired tokens are handled properly.
- **Project Management APIs:**
    + **Functionality**: APIs provide information about projects, including members, modules, releases, test plans, test cases, test runs, and issues.
    + **Verification**: Test cases verify the correct retrieval of project information, ensuring data integrity and proper access controls.
- **API Security:**
    + **Functionality**: The implementation includes user registration, login, logout, and role-based permissions.
    + **Verification**: Testing ensures that all relevant endpoints are protected and that unauthorized access is correctly blocked.

- **Front-End Development:**
  + **Functionality**: The system displays error pages (404, 400, 500) for incorrect paths or server issues.
  + **Verification**: Test cases ensure that users are redirected to the appropriate error pages under different failure scenarios.
- **Project List Page:**
  + **Functionality**: Displays a list of projects with pagination, search, and filter options. Users can create and delete projects directly from this page.
  + **Verification**: Tests verify the correct display and functionality of pagination, search, and filter features, as well as the creation and deletion of projects.
- **Dashboard Page:**
  + **Functionality**: Allows for user assignment to projects with specific roles (manager, tester, developer) and displays a history of actions performed within the project.
  + **Verification**: Test cases ensure that users are correctly assigned roles and that the history accurately reflects project activities.
- **Requirements, Modules, Releases, Test Cases, and Test Plans Pages:**
  + **Functionality**: Users can create, update, and delete requirements, modules, releases, test cases, and test plans, with each item having confirmation prompts for deletion.
  + **Verification**: Test cases confirm the correct creation, updating, and deletion of these elements, ensuring they are saved to the database and displayed correctly in the UI.

2. ***Out of Scope:***
3. ***Items not tested:***

- Not perform performance testing

IV. Metrics

1. *No. of test cases planned vs executed*

Build 1:

	Test Coverage:	**89%**								
	Successful Test Coverage:	**81%**							***Date:***	*2024/06/16*
No	**Requirement ID**	**Requirement name**	**Tested**	**Passed**	**Failed**	**Blocked**	**Skipped**	**Not Yet Tested**	**Total**	**Tested Coverage**
1	R001	Register	10	9	1	0	1	1	**12**	83%
2	R002	Login	7	7	0	0	0	1	**8**	88%
3	R003	Token provision	3	3	0	0	0	0	**3**	100%
4	R004	Token refreshing	4	3	1	0	0	0	**4**	100%
5	R005	API endpoint authorization	4	3	1	0	0	0	**4**	100%
6	R006	Project	23	9	14	0	0	0	**23**	100%
7	R007	Log out	2	2	0	0	3	0	**5**	40%
		Total	**24**	**22**	**2**	**0**	**1**	**2**	**27**	**89%**

Build 2:

	Test Coverage:	**90%**								
	Successful Test Coverage:	**47%**							***Date:***	*2024/06/16*
No	**Requirement ID**	**Requirement name**	**Tested**	**Passed**	**Failed**	**Blocked**	**Skipped**	**Not Yet Tested**	**Total**	**Tested Coverage**
1	R001	Register	1	1	0	0	1	1	**3**	33%
2	R002	Login	0	0	0	0	0	1	**1**	0%
3	R003	Token provision	3	3	0	0	0	0	**3**	100%

4	R004	Token refreshing	3	3	0	0	1	0	4	75%
5	R005	API endpoint authorization	3	3	0	0	1	0	4	75%
6	R006	Project	23	9	14	0	0	0	23	100%
7	R007	Log out	0	0	0	0	3	0	3	0%
8	R008	API - Project	15	7	8	0	3	0	18	83%
9	R009	API - Release	8	4	4	0	0	0	8	100%
10	R010	Test Error page	1	1	0	0	0	0	1	100%
11	R011	Test Projects page	8	8	0	0	0	0	8	100%
12	R012	API - Requirement	6	2	3	1	0	0	6	100%
13	R013	API - Module	6	2	4	0	0	0	6	100%
14	R014	Dashboard page	4	3	1	0	0	0	4	100%
15	R015	Requirement page	5	4	1	0	0	0	5	100%
16	R016	Test Api TestCase	5	0	1	4	0	0	5	100%
17	R017	Test Api TestPlan	5	0	1	4	0	0	5	100%
18	R018	Release Page	2	2	0	0	0	0	2	100%
19	R019	Test Case Page	1	0	1	0	0	0	1	100%
20	R020	Test Plan Page	1	0	1	0	0	0	1	100%
		Total	**100**	**52**	**39**	**9**	**9**	**2**	**111**	**90%**

Build Final:

	Test Coverage:	**98%**								
	Successful Test	**75%**							***Date:***	*<yyyy/mm/*

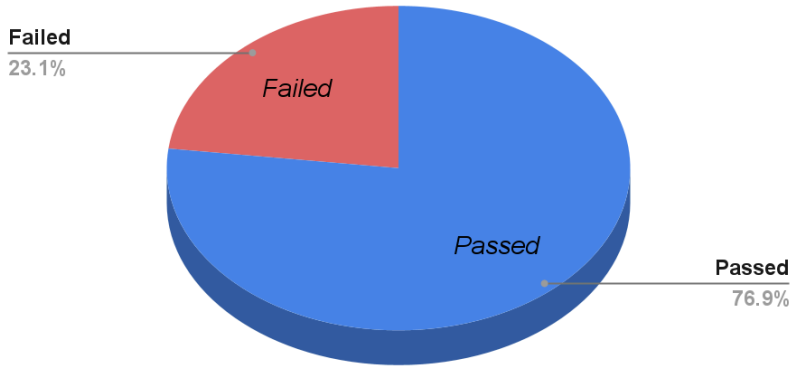
	Coverage:									*dd>*
No	**Requirement ID**	**Requirement name**	**Tested**	**Passed**	**Failed**	**Blocked**	**Skipped**	**Not Yet Tested**	**Total**	**Tested Coverage**
1	R001	API - Add Test Case	10	1	9	0	0	0	10	100%
2	R002	API - Edit Test Case	10	6	4	0	0	0	10	100%
3	R003	API - Add Test Run	10	10	0	0	0	0	10	100%
4	R004	State - Issues	13	13	0	0	0	0	13	100%
5	R005	Domain - Add Test Run	10	10	0	0	0	0	10	100%
6	R006	Domain - Add Issue	14	14	0	0	0	0	14	100%
7	R007	Domain - Add Comment	10	10	0	0	0	0	10	100%
8	R008	Domain - Users - Register	13	11	2	0	0	0	13	100%
9	R009	Domain - Projects - Add / Edit	5	5	0	0	0	0	5	100%
10	R010	Domain - Requirements - Add	8	0	8	0	0	0	8	100%
11	R011	Domain - Requirements - Edit	8	5	3	0	0	0	8	100%
12	R012	State - Register	11	11	0	0	0	0	11	100%
13	R013	API - Add Projects	9	4	5	0	1	0	10	90%
14	R014	API - Add Requirement	8	4	4	0	2	0	10	80%
15	R015	API - Edit Requirement	8	4	4	0	2	0	10	80%
16	R016	API - Add Release	10	9	1	0	0	0	10	100%
17	R017	API - Edit	10	6	4	0	0	0	10	100%

		Release								
18	R018	API - Add Test Plan	10	7	3	0	0	0	10	100%
19	R019	State - Release	12	12	0	0	0	0	12	100%
20	R020	Domain - Test Plan - Add	11	8	3	0	0	0	11	100%
21	R021	Domain - Release - Add	11	11	0	0	0	0	11	100%
22	R022	Domain - Module - Add	10	9	1	0	0	0	10	100%
		Total	**221**	**170**	**51**	**0**	**5**	**0**	**226**	**98%**

2. ***No. of test cases passed/failed***

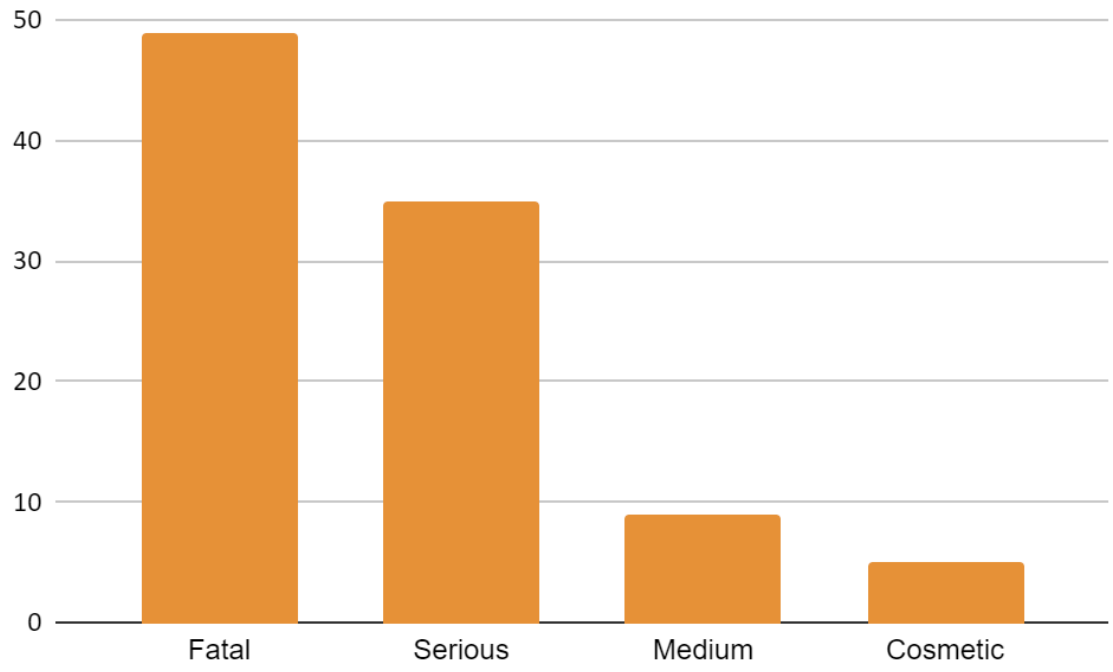
Build Final:

Test cases planned	Test cases executed	TCs Pass	Tcs Failed
226	221	170	51



Severity:

No.	Items	Fatal	Serious	Medium	Cosmetic	Total
1	API - Add Test Case	23	15	5	3	**46**
2	API - Edit Test Case	26	20	4	2	**52**
3	API - Add Test Run	0				**0**
4	State - Issues	0				**0**
5	Domain - Add Test Run	0				**0**
6	Domain - Add Issue					**0**
7	Domain - Add Comment					**0**
8	Domain - Users - Register	0	2	0	0	**2**
9	Domain - Projects - Add / Edit	0	0	0	0	**0**
10	Domain - Requirements - Add	0	0	8	0	**8**
11	Domain - Requirements - Edit	3	1	0	0	**4**
12	State - Register	0	0	0	0	**0**
13	API - Add Projects	3	1	1	0	**5**
14	API - Add Requirement	2	2	0	0	**4**
15	API - Edit Requirement	2	2	0	0	**4**
16	API - Add Release					**0**
17	API - Edit Release					**0**
18	API - Add Test Plan					**0**
19	State - Release					**0**
20	Domain - Test Plan - Add					**0**
21	Domain - Release - Add					**0**
22	Domain - Module - Add					**0**
	Total	**49**	**35**	**9**	**5**	**98**



V. Types of testing performed:

1. *Unit Testing*

- Scope: Unit Testing was conducted by the developers to verify the functionality of individual components or units of the Test Management website's back-end and front-end code.
- Focus: The primary objective was to ensure that each unit of the software, such as API endpoints, data processing functions, and user interface components, performed as expected in isolation. This involved testing functions, methods, classes, and state management logic.
- Methodology: Developers manually created and executed test cases for each unit, focusing on correct input handling, output validation, and error conditions, ensuring each component functioned correctly before integration.

2. ***Smoke Testing***
    - Scope: Smoke Testing was performed each time a new build (Release 1 or Release 2) was deployed into the test environment.
    - Focus: The goal was to ensure that the major functionalities of the application, such as user registration, login, project management, and test case management, were operational. This initial testing confirmed the build's stability and its readiness for more comprehensive testing.
3. ***System Integration Testing***
    - Scope: System Integration Testing was conducted to verify that the entire Test Management website met the specified requirements when integrated as a whole.
    - Focus: Testing centered on critical business scenarios, such as creating and managing test plans, assigning users to projects, and handling role-based permissions, ensuring that all integrated components worked together without errors.
4. ***Regression Testing***
    - Scope: Regression Testing was performed after each new build (containing bug fixes or new features) was deployed to ensure that existing functionalities continued to work correctly.
    - Coverage: This testing was extensive, covering the entire application, including both new functionalities and previously tested features, to detect any unintended side effects caused by the changes.
    - Purpose: The goal was to ensure that existing functionalities continued to work correctly after defect fixes and enhancements were integrated.
    - Methodology: The existing test suite was updated with new test cases related to the recent changes. These were executed alongside the existing test cases to ensure the system's stability and that no previously functioning features were broken.
    - TTest Environment & Tools:

Application URL	https://testingproject-ff89.onrender.com

VI. Lessons Learnt

No	Issues faced	Solutions
1	Manual Execution of Repetitive Tests	Allocated dedicated time and resources for testers to execute smoke tests manually, ensuring consistency and accuracy in the results.
2	Some test cases had unclear or ambiguous steps, resulting in inconsistent test results and confusion during execution.	Revised and updated the test cases to include clear, detailed steps and precise validation criteria, improving the reliability of test execution and outcomes.
3	Ensuring that all test cases were executed as required.	Implemented a thorough tracking system to monitor the execution status of all test cases, ensuring 100% execution.
4	Ensuring previously fixed defects did not recur.	Introduced a regression suite specifically for previously fixed defects, which was run in every regression cycle to verify that fixes held.
5	Completing and reviewing all required documentation.	Established a documentation review schedule and checklist to ensure that all required documents were complete, accurate, and reviewed on time.Implemented a thorough tracking system to monitor the execution status of all test cases, ensuring 100% execution.

VII. Recommendations

- Automate Repetitive Test Cases: Automation will enable faster feedback on the stability of new releases and reduce the manual effort required for each testing cycle.
- Enhance Test Case Documentation: Improve the clarity, detail, and precision in test case documentation. This will help avoid any ambiguities and ensure that all team members can execute the tests consistently and accurately.

- Comprehensive Regression Testing: Strengthen regression testing by developing a comprehensive regression suite that includes both new functionalities and previously fixed defects. This will help in ensuring that fixes hold and no new critical issues emerge.
- Documentation Review Checklist: Implement a documentation review checklist and schedule to ensure all required documents are complete, accurate, and reviewed on time. This will help in maintaining high-quality documentation throughout the project.
- Implement a Structured Documentation Review Process: Establish a documentation review checklist and schedule. This will ensure that all required documents, including test cases, test reports, and bug reports, are complete, accurate, and reviewed on time, maintaining high-quality standards throughout the project.

VIII. Best Practices

- Daily Stand-Up Meetings: Conduct daily stand-up meetings to facilitate clear communication among team members and quickly resolve any blockers. This practice helps keep the team aligned, ensures everyone is aware of their responsibilities, and allows for swift identification and resolution of issues.
- Clear and Detailed Documentation: All test cases, test plans, and test reports were documented clearly and in detail. This practice minimized ambiguities and ensured that all team members could execute tests consistently and understand the testing outcomes.
- Effective Defect Management and Prioritization: Implement a structured defect management and prioritization process. This ensures that critical and high-priority defects are addressed promptly, while less critical issues are tracked with a clear action plan for resolution, leading to more efficient and effective testing cycles.

IX. Exit Criteria

1. All test case should be executed **- Yes**
2. 80% of all test cases must pass **- Yes**

3. All critical and high-priority defects must be resolved and closed **- Yes**
4. No critical or high-priority defects should remain open **- Yes**
5. Completed regression testing with no new critical or high-priority defects **- Yes**
6. Ensured that previously fixed defects have not recurred **- Yes**
7. All required documentation (e.g., test plans, test cases, test summary reports) is complete and reviewed. **- Yes**

X. Conclusion/Sign Off

After thorough testing of the Test management website project, including Release 1, Release 2 and Final, and satisfying all the defined Exit Criteria, the Testing team concludes the following: **Exit Criteria**

Given that the Exit Criteria have been met and all testing objectives have been achieved, the Testing team recommends that the application be approved for the 'Go Live' phase. However, it is advised that appropriate User/Business Acceptance Testing (UAT) be conducted before the final deployment to ensure alignment with user expectations and business requirements.

XI. Definitions, Acronyms, and Abbreviations

1. Definitions:

- **Test Case:** A set of conditions or variables under which a tester assesses whether an application or system is working as intended.
- **Defect:** An issue or bug found during testing that indicates a deviation from the expected behavior or requirements.
- **Build:** A version of the software that is compiled and deployed for testing.
- **Regression Testing:** Testing conducted to confirm that recent changes or fixes have not adversely affected existing functionalities.
- **Smoke Testing:** A preliminary test to check the basic functionality of a build before detailed testing.

2. Acronyms:

- **UAT (User Acceptance Testing):** Testing performed by end-users to validate if the application meets their needs and requirements.

- **TC (Test Case):** A set of conditions or variables for testing an application or system.
- **SRS (Software Requirements Specification):** A document detailing the required functionalities and features of a software application.
- **HP QC/ALM (Hewlett Packard Quality Center/Application Lifecycle Management):** A tool for managing the software testing process, including test planning, execution, and defect tracking.
- **API (Application Programming Interface):** A set of rules and protocols for building and interacting with software applications.
- **PM (Project Manager):** The person responsible for planning, executing, and closing the project.

3. ***Abbreviations:***

- **Build 1:** The first version of the application released for testing.
- **Build 2:** The second version of the application released for testing, incorporating additional features and fixes.
- **Postgres:** A powerful, open-source relational database management system used in the project.
- **Docker:** A platform used to develop, ship, and run applications in containers.
- **TCs (Test Cases):** Individual scenarios designed to test specific aspects of the application.

4. ***Metrics:***

- **Test Coverage:** The percentage of the total number of test cases executed versus the total number planned.
- **Successful Test Coverage:** The percentage of test cases that passed successfully.

5. ***Severity Levels:***

- **Immediate:** Critical issues that need to be addressed immediately to ensure the application functions correctly.
- **Urgent:** High-priority issues that need to be resolved promptly but are not critical.

- **High:** Significant issues that impact the application's functionality but do not halt its overall operation.
- **Normal:** Standard issues that do not significantly impact the application's functionality.
- **Low:** Minor issues that have minimal impact on the application's operation.

WORK ASSIGNMENT

No	Task	Assignee	Deadline
PHASE 1:			
0	Perform testing of the 'Register,Login, Logout ' function. (Build1)	Thủy Tiên	00:00 17/06/2024
1	Perform testing of the 'Token provision, Token refreshing, API endpoint authorization ' function. (Build1)	Văn Minh	00:00 17/06/2024
2	Perform testing of the 'Project ' function. (Build1)	Hải Nam	00:00 17/06/2024
3	Compile and review Test Cases for Build 1.	Văn Minh	00:00 18/06/2024
4	Create a TEST PLAN template. **(Requirement 1**)	Thủy Tiên	00:00 10/06/2024
5	TEST PLAN **(Requirement 1)**	Thủy Tiên, Văn Minh, Hải Nam	00:00 17/06/2024
PHASE 2:			
6	Perform testing of the ‘Register, Login, Log out, API - Project, API - Release' function. (Build2)	Thủy Tiên	00:00 13/07/2024
7	Perform testing of the 'Token provision, Token refreshing, API endpoint authorization, API - Requirement, API - Module, Dashboard page,	Văn Minh	00:00 13/07/2024

	Requirement page' function. (Build2)		
8	Perform testing of the 'Project, Test Error page, Test Projects page, Test Api TestCase, Test Api TestPlan, Release PageTest Case Page ' function. (Build2)	Hải Nam	00:00 13/07/2024
9	Compile and review Test Cases for Build 2.	Văn Minh	00:00 14/07/2024
PHASE 3:			
10	Create a TEST SUMMARY REPORT template.	Thủy Tiên	00:00 20/07/2024
11	**Requirement 2: Domain Testing** ● Perform testing of the 'Users (Register, Login, Logout)' function. (Build Final) ● Perform testing of the 'Projects (Add, Edit, Delete Project)' function. (Build Final) ● Perform testing of the 'Requirements(Add, Edit, Delete Requirement)' function. (Build Final)	Thủy Tiên	00:00 25/08/2024
12	**Requirement 2: Domain Testing** ● Perform testing of the 'Test Plans (Add Test Plan)' function. (Build Final) ● Perform testing of the 'Releases (Add, Edit, Delete Release)' function. (Build Final) ● Perform testing of the 'Modules (Add, Edit, Delete Module)' function. (Build Final)	Văn Minh	00:00 25/08/2024

13	**Requirement 2: Domain Testing** ● Perform testing of the 'Test cases (Add, Edit, Delete TC (API))' function. (Build Final) ● Perform testing of the 'Test runs (Add, Edit, Delete TR)' function. (Build Final) ● Perform testing of the 'Issues/Bugs (Add, Edit, Delete Issue)' function. (Build Final)	Hải Nam	00:00 25/08/2024
14	**Requirement 3: State Transition Testing** ● Perform testing of the 'Users (Register) - (Build Final)	Thủy Tiên	00:00 25/08/2024
15	**Requirement 3: State Transition Testing** ● Perform testing of the Releases function. (Build Final)	Văn Minh	00:00 25/08/2024
16	**Requirement 3: State Transition Testing** ● Perform testing of the 'Issues/Bugs' (Build Final)	Hải Nam	00:00 25/08/2024
17	**Requirement 4: Automation testing** ● Perform testing of the 'Users (Register, Login, Logout)' function. (Build Final) ● Perform testing of the 'Projects (Add, Edit, Delete Project)' function. (Build Final) ● Perform testing of the 'Requirements(Add, Edit, Delete Requirement)' function. (Build Final)	Thủy Tiên	00:00 25/08/2024
18	**Requirement 4: Automation testing** ● Perform testing of the 'Test Plans (Add Test	Văn Minh	00:00 25/08/2024

	Plan)' function. (Build Final) ● Perform testing of the 'Releases (Add, Edit, Delete Release)' function. (Build Final) ● Perform testing of the 'Modules (Add, Edit, Delete Module)' function. (Build Final)		
19	**Requirement 4: Automation testing** ● Perform testing of the 'Test cases (Add, Edit, Delete TC (API))' function. (Build Final) ● Perform testing of the 'Test runs (Add, Edit, Delete TR)' function. (Build Final) ● Perform testing of the 'Issues/Bugs (Add, Edit, Delete Issue)' function. (Build Final)	Hải Nam	00:00 25/08/2024
20	**Requirement 5: GUI testing** ● Perform testing of the 'Users (Register, Login, Logout)' UI . (Build Final) ● Perform testing of the 'Projects (Add, Edit, Delete Project)' UI. (Build Final) ● Perform testing of the 'Requirements(Add, Edit, Delete Requirement)' UI. (Build Final)	Thủy Tiên	00:00 25/08/2024
21	**Requirement 5: GUI testing** ● Perform testing of the 'Test Plans (Add Test Plan)' UI. (Build Final) ● Perform testing of the 'Releases (Add, Edit, Delete Release)' UI. (Build Final) ● Perform testing of the 'Modules (Add, Edit, Delete Module)' UI. (Build Final)	Văn Minh	00:00 25/08/2024

22	**Requirement 5: GUI testing** ● Perform testing of the 'Test cases (Add, Edit, Delete TC (API))' UI. (Build Final) ● Perform testing of the 'Test runs (Add, Edit, Delete TR)' UI. (Build Final) ● Perform testing of the 'Issues/Bugs (Add, Edit, Delete Issue)' UI. (Build Final)	Hải Nam	00:00 25/08/2024
23	**Requirement 6: Performance testing**	Thủy Tiên, Văn Minh, Hải Nam	00:00 25/08/2024
24	**Requirement 7: API testing** ● Perform testing of the 'Users (Register, Login, Logout)' function. (Build Final) ● Perform testing of the 'Projects (Add, Edit, Delete Project)' function. (Build Final) ● Perform testing of the 'Requirements(Add, Edit, Delete Requirement)' function. (Build Final)	Thủy Tiên	00:00 25/08/2024
25	**Requirement 7: API testing** ● Perform testing of the 'Test Plans (Add Test Plan)' function. (Build Final) ● Perform testing of the 'Releases (Add, Edit, Delete Release)' function. (Build Final) ● Perform testing of the 'Modules (Add, Edit, Delete Module)' function. (Build Final)	Văn Minh	00:00 25/08/2024
26	**Requirement 7: API testing**	Hải Nam	00:00

	• Perform testing of the 'Users (Register, Login, Logout)' function. (Build Final) • Perform testing of the 'Projects (Add, Edit, Delete Project)' function. (Build Final) • Perform testing of the 'Requirements(Add, Edit, Delete Requirement)' function. (Build Final)		25/08/2024
27	**Requirement 8: Data generation**	Thủy Tiên, Văn Minh, Hải Nam	00:00 25/08/2024
28	**Requirement 9: Test case tracking system & Bug tracking system**	Thủy Tiên, Văn Minh, Hải Nam	00:00 27/08/2024
29	Compile and review Test Cases for Build final.	Văn Minh	00:00 27/08/2024
30	Test Report	Thủy Tiên, Văn Minh, Hải Nam	00:00 27/08/2024
31	TEST SUMMARY REPORT	Thủy Tiên,	00:00 22/08/2024
32	Compile and review Test Cases for TEST SUMMARY REPORT.	Thủy Tiên	12:00 26/08/2024

MEMBER EVALUATION

Group: 11

ID	Name	Completion	Comments
21127360	Phạm Văn Minh	90%	Not perform performance testing
21127365	Phan Phước Hải Nam	90%	Not perform performance testing
21127699	Lô Thủy Tiên	90%	Not perform performance testing

REFERENCES

Supporting Tools and Software:

No	Function	Tool
[1]	System	Github, VScode
[2]	Web	Google, Opera
[3]	Testing	Katalon, Jmeter, PostMan. Mockaroo
[4]	Project Management	Discord, Google drive, Messenger

References:

[0] Software Testing course materials - 21KTPM2

[1] *Sample-Test-Summary-Report*, by-SoftwareTestingHelp, link